# Collections Copilot

Adda Wale | Implemented local build | Snapshot 28 September 2026

Source release: CRM, cards, lending, core banking, collections and bureau data. CSV / Parquet, bilingual notes/transcripts, current policies and optional synthetic WAV audio.

## 1 Data Product Factory

DuckDB raw → curated → C360
Typed checks retain raw values and string IDs.
Cross-source identity rules; uncertain links held for review, not automatically merged.
Outputs: C360, lineage, counted quality report and YAML contract.

## 2 Insight / NLP

FastAPI + reviewed metric compiler
Validated local / optional Gemini plans → bound SQL.
Policy retrieval and exact transcript statements.
Optional verified WAV playback + local Whisper ASR.
Outputs: cited answers, benchmark CSV and review-only voice evidence.

## 3 Feature Store

One versioned offline/live definition
Structured features and bilingual financial cues.
100-customer parity: zero mismatches.
ASR cache reuses cues; speakers remain unknown.
Outputs: stored features and source/version registry; voice cues do not enter the classifier.

## 4 Model / Decisioning

AI: scikit-learn logistic review classifier
Rules: holds → recorded hardship → possible cues.
Facts, reasons and draft; optional Gemini evidence selection cannot change decisions.
Human: supervisor approve / reject / edit.
Outputs: pending brief and SQLite review/audit.

Application: Next.js / Tailwind / Radix → FastAPI. Local DuckDB analytics; SQLite sessions/reviews. Gemini is optional and stateless; no key means local operation.

Governance across layers: server-issued roles; assigned-case access; source lineage and audit; protected fields excluded from serving/features. No automatic contact. Match scores are uncalibrated.